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EXP – 08: BANKERS DEADLOCK AVOIDANCE ALGORITHM

```
user123@localhost:~/bankers
[user123@localhost ~]$ ls
abc.sh  awk     Desktop   Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers Documents Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd bankers
[user123@localhost bankers]$ vi bankers.c
[user123@localhost bankers]$
```

```
user123@localhost:~/bankers — /usr/bin/vim bankers.c
#include <stdio.h>

int main(){
        int n, r, i, j, k;

        printf("Enter number of processes: ");
        scanf("%d", &n);

        printf("Enter number of resource types: ");
        scanf("%d", &r);

        int alloc[n][r], max[n][r], need[n][r], avail[r];
        int finish[n], safe_sequence[n];
        int count=0, safeFound;

        //Allocation Matrix
        printf("Enter Allocation Matrix:\n");
        for(int i=0; i<n; i++){
                for(int j=0; j<r; j++){
                        scanf("%d",&alloc[i][j]);
                }
        }

        //Max Matrix
        printf("Enter Max Matrix:\n");
        for(i=0; i<n; i++){
                for(j=0; j<r; j++){
                        scanf("%d",&max[i][j]);
                }
        }

        //Available Matrix
        printf("Enter Available Resources:\n");
        for(i=0; i<r; i++){
                scanf("%d", &avail[i]);
        }

        //Initialize finish array
        for(int i=0; i<n; i++){
                finish[i]=0;
        }

        //Calculate Need Matrix
        for(i=0; i<n; i++){
                for(j=0; j<r; j++){
                        need[i][j] = max[i][j] - alloc[i][j];
                }
        }

        //Bankers Algorithm
        while(count < n){
                safeFound=0;

                for(int i=0; i<n; i++){
                        if(finish[i]==0){
                                int safe = 1;
                                for(j=0; j<r; j++){
                                        if(need[i][j]>avail[j]){
                                                safe = 0;
                                                break;
                                        }
                                }

                                if(safe){
                                        for(k=0; k<r; k++){
                                                avail[k] = avail[k] + alloc[i][k];
                                        }
                                        safe_sequence[count] = i;
                                        finish[i] = 1;
                                        safeFound = 1;
                                        count++;
                                }
                        }
                }

                if(!safeFound){
                        printf("System is in unsafe state! May occur the deadlock \n");
                        return 0;
                }
        }
```

```c
        //print safe sequence
        printf("Safe sequence is: ");
        for(i=0; i<n; i++){
                printf("P%d ",safe_sequence[i]);
        }

        return 0;
}
```

```
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[user123@localhost ~]$ ls
abc.sh  awk      Desktop    Downloads  ifelse.sh  memory  page  Pictures   Public   REC       sample.sh   scheduling  SFA  system_calls  Videos
ab.sh   bankers  Documents  Folder     loops.sh   Music   pc    posneg.sh  qns.txt  sample.c  sample.txt  script      sh   Templates     while.sh
[user123@localhost ~]$ cd bankers
[user123@localhost bankers]$ vi bankers.c
[user123@localhost bankers]$ gcc bankers.c
[user123@localhost bankers]$ ./a.out
Enter number of processes: 4
Enter number of resource types: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 0
Enter Max Matrix:
7 5 3
3 2 2
7 0 2
2 2 2
Enter Available Resources:
3 3 2
Safe sequence is: P1 P2 P3 P0 [user123@localhost bankers]$
```